MATH73235 Spring 2025 Quiz 8

First name:

Last name:

Student ID:

Name and Student ID must be filled in for this page to be graded.

Preamble: You may use any result from the "Helpful Results" section.

Problem 1. *Use Fourier Series to obtain a standing wave solution $u(x,t)$ to the 1D wave-equation for a string of length $L = 1$ and $c^2 = 1$, when the initial velocity is zero, and the initial deflection is given by* $f(x) = 0.01\left(\sin(\pi x) - \frac{1}{2}\sin(2\pi x)\right)$.

[5 marks]

Solution. We have

$$u(x,0) = \sum_{n=1}^{\infty} A_n \sin(n\pi x) = f(x) \quad (\text{as } L=1)$$

and $A_n = 2\int_0^1 f(x)\sin(n\pi x)\,dx$

Then

$$A_1 = 2\int_0^1 0.01\left(\sin(\pi x) - \frac{1}{2}\sin(2\pi x)\right)\cdot \sin(\pi x)\,dx$$

$$= 0.01\left[\underbrace{2\int_0^1 \sin(\pi x)\sin(\pi x)\,dx}_{=1} - \underbrace{\int_0^1 \sin(2\pi x)\sin(\pi x)\,dx}_{=0}\right]$$

$\Rightarrow A_1 = 0.01$

Similarly: $A_2 = 0.01\left(-\frac{1}{2}\right) = -0.005$

and $A_3 = A_4 = \dots = A_n = \dots = 0$

Thus

$$u(x,t) = 0.01\sin(\pi x)\cos(\pi t) - 0.005\sin(2\pi x)\cos(2\pi t). \quad \square$$

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Problem 2. *Write down the D'Alembert's solution $u(x,t)$ to the 1D wave-equation with initial conditions $u(x,0) = f(x)$ and $u_t(x,0) = 0$ when*

$$f(x) = 0.01x(1-x).$$

[5 marks]

Solution.

D'Alembert's soln:

$$u(x,t) = \frac{1}{2}\left[f(x-ct) + f(x+ct)\right]$$

Here $f(x) = 0.01x(1-x)$, so

$$u(x,t) = \frac{1}{2}\left[0.01(x-ct)(1-(x-ct)) + 0.01(x+ct)(1-(x+ct))\right]$$

$$= \frac{1}{2}\left[0.01(x-ct)(1-x+ct) + 0.01(x+ct)(1-x-ct)\right]$$

$$= \frac{1}{2}(0.01)\left[x - \cancel{ct} - x^2 + \cancel{xct} + \cancel{xct} - c^2t^2 + x + \cancel{ct} - x^2 - \cancel{xct} - \cancel{xct} - c^2t^2\right]$$

$$= 0.005\left[2(x - x^2 - c^2t^2)\right] = 0.01(x - x^2 - c^2t^2). \square$$

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Extra Page: Must submit even if unused.

Helpful Results

- The one-dimensional wave equation $u_{tt} = c^2 u_{xx}$ with standing wave boundary conditions $u(0,t) = 0 = u(L,t)$ for $t \geq 0$ and $L > 0$, and initial conditions $u(x,0) = f(x)$, $u_t(x,0) = 0$ for $0 \leq x \leq L$, has solution based on Fourier Series given by

$$u(x,t) = \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi c}{L} t\right) \sin\left(\frac{n\pi}{L} x\right),$$

  with $A_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L} x\right)\, dx$.

- The one-dimensional wave equation $u_{tt} = c^2 u_{xx}$ with initial conditions $u(x,0) = f(x)$, $u_t(x,0) = 0$ for $-\infty < x < \infty$, and no specific boundary conditions, has solution based on D'Alembert's work:

$$u(x,t) = \frac{1}{2}\left[f(x-ct) + f(x+ct)\right].$$